



杰曼诺夫数学中心
Shenzhen International Center for Mathematics

International Conference on Symplectic Dynamics August 25-29, 2025



August 25-29
2025



The Shenzhen International Center for Mathematics, officially established on February 22, 2019, is a scientific research center funded by the Shenzhen Municipal Government and located in Southern University of Science and Technology. The scientific director is Fields Medalist Efim Zelmanov.

It aims to advance scientific research in pure mathematics, applied and computational mathematics in Shenzhen, the Greater Bay Area, and beyond.

The center will focus on important research fields in both pure and applied mathematics.

Its primary goal is to promote the international scientific cooperation and advance research in mathematics and interdisciplinary fields of life science, information science, engineering, finance, etc.

The SICM works in accordance with international standards for major international mathematics centers. It provides first-class research environments for researchers and visitors.

Hosting annual thematic programs, conferences, and workshops.

**Welcome to the Shenzhen
International Center for Mathematics!**



CONFERENCE GUIDE

Wifi: ICMPub

Password: ICM88888

REGISTRATION

August 25, 9:00-9:30AM SICM 240A, Taizhou Building Second Floor.

南方科技大学深圳国际数学中心报告厅(台州楼240A)

ACCOMODATION

★ **Shenzhen Jinfeng Hotel, 4168-B Liu Xian Avenue, Nanshan District, Shenzhen.**

Phone: 0755-86681088.

深圳市南山区留仙大道4168-B号, 深圳璟峯酒店(西丽大学城店)

★ **Yayuan Tanglang Hotel, Tianliao Building, Xueyuan Avenue, Xili Town, Shenzhen, Nanshan District**

深圳市南山区桃源街道大学城学苑大道田寮大厦1113号, 雅园塘朗酒店

Campus Map



地图 map
南方科技大学
southern university of science and technology

Schedule

	Mon.25-08	Tue.26-08	Wed.27-08	Thu.28-08	Fri.29-08
9:00 - 9:50	Jeff Xia	Urs Frauenfelder	Leonardo Macarini	Xijun Hu	Marco Mazzucchelli
10:00 - 10:50	Alberto Abbondandolo	Otto van Koert	Gabriele Benedetti	Jungsoo Kang	Jean Gutt
10:50 - 11:20	Coffee Break 				
11.20 - 12:10	Umberto Hryniewicz	Igor Uljarevic	Yu-Wei Fan	Oliver Edtmair	Jingling Yang
12:10 - 14:30	Lunch 				
14:30 - 15:20	Yu Pan	Julian Chaidez	Discussing	Lei Liu	Brayan Ferreira
15:30 - 16:20	Matthias Meiwes	Marta Batoreo		Dustin Connery-Grigg	Yuan Gao
16:20 - 16:50	Coffee Break 				
16:50 - 17:40	Luya Wang	Han Lou	Discussing	Lucas Dahinden	Marcelo Atallah
18:30 - 20:30	Dinner 			Dinner 	

Titles and abstracts:

Length spectrum rigidity and flexibility of spheres of revolution

Alberto Abbondandolo

How much information is encoded by the periodic orbits of a dynamical system? I will discuss this question in the setting of geodesic flows on S^1 -symmetric Riemannian two-spheres having only one equator. This talk is based on a joint work with Marco Mazzucchelli.

C^0 -rigidity of the Hamiltonian diffeomorphism group of symplectic rational surfaces

Marcelo Atallah

A natural question bridging the celebrated Gromov–Eliashberg theorem and the C^0 -flux conjecture is whether the identity component of the group of symplectic diffeomorphisms is C^0 -closed in $\text{Symp}(M, \omega)$. Beyond surfaces and the cases in which the Torelli subgroup of $\text{Symp}(M, \omega)$ coincides with the identity component, little is known. In joint work with Cheuk Yu Mak and Wewei Wu, we show that, for all but a few positive rational surfaces, the group of Hamiltonian diffeomorphisms is the C^0 -connected component of the identity in $\text{Symp}(M, \omega)$, thereby giving a positive answer in this setting. Here, “positive rational surface” essentially means a k -point blow-up of $\mathbb{CP}^2$ whose symplectic form evaluates positively on the first Chern class.

The number of periodic points of surface symplectomorphisms

Marta Batoreo

In this talk I will survey some results on the existence of periodic points of symplectomorphisms defined on closed orientable surfaces of positive genus g . Namely, I will describe some symplectic flows on such surfaces possessing finitely many periodic points and describe a non-Hamiltonian variant of the Hofer-Zehnder conjecture for symplectomorphisms defined on surfaces; this conjecture provides a quantitative threshold on the number of fixed points (possibly counted homologically) which forces the existence of infinitely many periodic points. This is joint work in progress with Marcelo Atallah and Brayan Ferreira.

Rigidity and flexibility of periodic Hamiltonian systems

Gabriele Benedetti

An old problem in classical mechanics asks for the existence of Hamiltonian systems (for instance within the class of central forces or geodesic flows) whose orbits are all periodic irrespective of their initial condition. Such problem has gained new life in recent years since these periodic systems are local maxima for systolic inequalities in symplectic geometry. While Bertrand (1873) showed that only trivial examples of periodic systems exist among central forces (the gravitational and elastic potential), Zoll (1903) and, later, Guillemin (1976) proved that there are many exotic examples among geodesic flows on the two-sphere. Following Guillemin's approach, recently extended by Ambrozio, C. Marques and Neves (2021) to higher dimensions, the goal of this talk is to construct magnetic flows (a generalization of geodesic flows in which the particle is subject to a Lorentz force) on the two-torus which are periodic for just one single value of the energy. This is joint work with Luca Asselle and Massimiliano Berti.

Creased contact manifolds and billiard dynamics

Julian Chaidez

Billiard dynamics is an old subject with many landmark theorems going back to Birkhoff. However, while billiard dynamics on tables with convex boundary is relatively well understood, basic questions remain open in the non-convex case. For instance, Arnaud has asked if every smooth (possibly non-convex) billiard table has infinitely many simple closed billiard orbits.

In this talk, I will explain a framework for studying billiards on any Riemannian (or Finsler) manifold using tools from contact geometry (e.g. spectral invariants coming from Floer theory). Specifically, I will introduce a type of singular contact form, called a creased contact form, along with an accompanying notion of generalized Reeb flow. This is the natural structure on the boundary of the unit cotangent bundle of a manifold with boundary, which corresponds to the billiard case. I will also explain the natural way to associate a linearized flow and Conley-Zehnder index to closed Reeb orbits in this setting. I will discuss various applications towards Arnaud's question, and ongoing work towards the full solution.

This is work in preparation joint with Shira Tanny (Weizmann).

Dynamic and topological aspects of Hamiltonian Floer theory on surfaces

Dustin Connery-Grigg

Given a Hamiltonian $H \in C^\infty(S^1 \times M)$, what is the relationship between the dynamics of the isotopy generated by H and the various Floer-theoretic invariants associated to H ? In this talk I will discuss how — in the case where M is a closed surface — one can interpret certain spectral invariants in purely dynamical terms, in addition to interpreting certain parts of the Hamiltonian Floer complex in terms of particularly nice transverse foliations initially introduced by Le Calvez in his study of equivariant Brouwer theory and surface homeomorphisms.

Self-intersections of Finsler and Reeb orbits

Lucas Dahinden

It is well known that two distinct geodesics in a compact Riemannian manifold cannot intersect infinitely often in finite time. This statement becomes wrong in general if one replaces Riemannian dynamics by non-reversible Finsler or even Reeb dynamics, essentially because Finsler and Reeb orbits can be tangent (in the base manifold) without coinciding. Nonetheless, we show that the statement generically holds. We then prove that for generic contact forms in dimension greater than 2, orbits joining two fixed points and closed orbits do not have self-intersections at all, generalizing a theorem by Rademacher. This is joint work with Jacobus de Pooter.

On the topological invariance of helicity

Oliver Edtmair

Helicity is an invariant of divergence free vector fields on a three-manifold. One of its fundamental properties is invariance under volume preserving diffeomorphisms. Arnold, having derived an ergodic interpretation of helicity as an asymptotic Hopf invariant, asked whether helicity remains invariant under volume preserving homeomorphisms, and more generally, whether it admits an extension to topological volume preserving flows. In this talk, I will present an affirmative answer to both questions for non-singular flows. The proof draws on recent advances in C^0 symplectic geometry, in particular regarding the algebraic structure of the group of area preserving homeomorphisms, which I will also survey. This is based on joint work with Sobhan Seyfaddini.

Finite subgroups of autoequivalences of general K3 surfaces

Yu-Wei Fan

We will describe a classification of finite subgroups of the group of autoequivalences of the bounded derived category of coherent sheaves on a general K3 surface. The main tool involves a categorical analogue of the Nielsen realization, in terms of autoequivalences acting on the space of Bridgeland stability conditions. Necessary background and motivation from mirror symmetry will also be discussed. Joint work with Kuan-Wen Lai.

Symplectic embeddings in T^*S^2

Brayan Ferreira

The question whether a symplectic manifold can be symplectically embedded into another is central in symplectic topology. Since Gromov's nonsqueezing theorem, we know that the problem is more subtle than its volume preserving counterpart. Examples in open subsets of $\mathbb{R}^{2n}$ already show that the question can be highly nontrivial. In dimension 4, specially for toric domains, the problem is substantially better understood - largely thanks to obstructions arising from embedded contact homology (ECH). In this talk we focus on symplectic embedding problems in which the target manifold is a disk cotangent bundle of a two-dimensional sphere, i.e., the set consisting of the covectors with norm less than a constant over a Riemannian sphere. We shall discuss tools such as symplectic capacities, action-angle coordinates, and possible applications to systolic inequalities. Much of this talk is based on joint works with Vinicius Ramos and Alejandro Vicente.

On EBK quantization

Urs Frauenfelder

This is joint work with Kai Cieliebak. EBK quantization is a quantization technique using quantized Arnold-Liouville tori. In the talk I discuss how for toric domains EBK quantization is related to the action spectrum of periodic orbits by a minimax procedure. This minimax procedure is reminiscent of spectral numbers in Floer homology. A strong candidate to explain these phenomena is Tate Rabinowitz homology for a delayed version of Rabinowitz action functional.

Rabinowitz Floer theory and categorical completion

Yuan Gao

Rabinowitz Floer homology has proven a powerful tool in studying topology and dynamics in both symplectic and contact contexts, but in many situation the lack of functoriality causes certain computational difficulty. In this talk, I will revisit some categorical constructions on obtaining formal completions, originally due to Efimov, and explain how this construction can yield an algebraic approach to extending the Rabinowitz Fukaya category on fully stopped Liouville manifolds. This construction is in turn related, via mirror symmetry, to objects on a formal neighborhood.

Ekeland-Hofer capacities from positive S^1 equivariant symplectic homology

Jean Gutt

Associated to a star-shaped domain in $\mathbb{R}^{2n}$ are two increasing sequences of capacities: the Ekeland-Hofer capacities and the so-called Gutt-Hutchings capacities. I shall recall both constructions and then present the main theorem that they are the same. This is joint work with Vinicius Ramos

Infinitely many chords, homoclinic orbits and nice Birkhoff sections for Reeb flows in 3D

Umberto Hryniewicz

The main result of this talk is the following statement: If a Reeb vector field on a closed 3-manifold admits a boundary strong Birkhoff section then every Legendrian knot has infinitely many geometrically distinct Reeb chords, except possibly when the ambient manifold is a lens space or the sphere and the Reeb flow has exactly two periodic orbits. This is joint work with Vincent Colin and Ana Rechtman.

A Symplectic Dynamics Approach to a Class of Spatial N-Body Problems

Xijun Hu

We study a class of N-body problems with essential three degrees of freedom and rotational symmetry. By reducing the rotational symmetry, we investigate periodic orbits in the resulting two-degree-of-freedom Hamiltonian system. For specific choices of angular momentum and energy, the dynamics on the energy surface corresponds to a Reeb flow on the tight three-sphere. We identify a Hopf link formed by a planar orbit and a symmetric brake orbit, which generates an open book decomposition with annulus-like global surfaces of section. Furthermore, we prove the existence of infinitely many rotationally distinct periodic orbits on some compact regular energy surface for both the spatial isosceles three-body problem and the Gutzwiller anisotropic Kepler problem. Applications to perturbed Kepler problems are also discussed.

This talk is based on joint works with Lei Liu, Yuwei Ou, Zhiwen Qiao, Pedro A. S. Salomão, and Guowei Yu

Floer homology of Legendrian submanifolds in prequantization bundles

Jungsoo Kang

Prequantization bundles are principal circle bundles over symplectic manifolds that carry natural contact structures. Certain monotone Lagrangian submanifolds in the base manifold admits Legendrian lifts to the total space of the bundle. In this talk, I will discuss how the Floer homology of a Lagrangian submanifold in the base is related to the Rabinowitz Floer homology of its Legendrian lift. This is joint work in progress with Hanwool Bae and Sungho Kim.

Finite energy foliations in the restricted three-body problem

Lei Liu

This talk is about using pseudo-holomorphic curves to study the circular planar restricted three-body problem. The main result states that for mass ratios sufficiently close to $1/2$ and energies slightly above the first Lagrange value, the flow on the regularized component $\mathbb{R}P^3 \setminus \mathbb{R}P^3$ of the energy surface admits a finite energy foliation with three binding orbits, namely two retrograde orbits around the primaries and the Lyapunov orbit in the neck region about the first Lagrange point. This foliation explains the numerically observed homoclinic orbits to the Lyapunov orbits. The critical energy surface is proved to satisfy the strict convexity condition in regularizing elliptic coordinates. This allows for the application of a general abstract result for Reeb vector fields on holed lens spaces, concerning the existence of finite energy foliations with prescribed binding orbits. As a by-product of the convex analysis, Birkhoff's retrograde orbit conjecture is proved for mass ratios sufficiently close to $1/2$ and all energies below the first Lagrange value. This conjecture states that the retrograde orbit bounds a disk-like global surface of section on each regularized component $\mathbb{R}P^3$ of the energy surface. This work is joint with Professor Pedro Salomao.

On Lagrangian Tori in $S^2 \times S^2$

Han Lou

In [FOOO12], K. Fukaya, Y. Oh, H. Ohta, and K. Ono (FOOO) obtained the monotone symplectic manifold $S^2 \times S^2$ by resolving the singularity of a toric degeneration of a Hirzebruch surface. They identified a continuum of toric fibers in the resolved toric degeneration that are not Hamiltonian isotopic to the toric fibers of the standard toric structure on $S^2 \times S^2$. In this paper, we provide a comprehensive classification: for any toric fiber in FOOO's construction of $S^2 \times S^2$, we determine whether it is Hamiltonian isotopic to a toric fiber of the standard toric structure of $S^2 \times S^2$.

Existence and localization of closed magnetic geodesics with low energy

Leonardo Macarini

Magnetic flows are generalizations of geodesic flows that describe the motion of a charged particle in a magnetic field. While every closed Riemannian manifold admits at least one closed geodesic, the analogous problem for magnetic orbits (also known as magnetic geodesics) is significantly more challenging and has received considerable attention in recent decades. I will present a result establishing that every low energy level of any magnetic flow admits at least one contractible closed orbit, assuming only that the magnetic strength is not identically zero, has a compact strict local maximum K , and that the cohomology class of the magnetic field is spherically rational. Moreover, this magnetic geodesic can be localized within an arbitrarily small neighborhood of K . This is joint work with Valerio Assenza and Gabriele Benedetti.

FROM CURVE SHORTENING TO FLAT LINK STABILITY AND BIRKHOFF SECTIONS OF GEODESIC FLOWS

Marco Mazzucchelli

In this talk, based on joint work with Marcelo Alves, I will present three new theorems on the dynamics of geodesic flows of closed Riemannian surfaces, proved using the curve shortening flow. The first result is the stability, under C^0 -small perturbations of the Riemannian metric, of certain flat links of closed geodesics. The second one is a forced existence theorem for closed geodesics on orientable closed Riemannian surfaces. The third theorem asserts the existence of Birkhoff sections for the geodesic flow of any closed orientable Riemannian surface.

Braids and spectral distance

Matthias Meiwes

Braids play an important role in surface dynamics. There is a strong connection between the existence of specific braids of periodic orbits, and central dynamical notions such as topological entropy, integrability, etc. There are so far several approaches to understand braids of periodic orbits of symplectic diffeomorphisms in connection with Floer theory.

In my talk, I would like to explain some persistence phenomenon of braids with respect to the distance on the group of Hamiltonian diffeomorphisms induced by a variant of the Oh-Schwarz spectral invariants, recently introduced by Connery-Grigg. A consequence will be the following: If the topological entropy of a Hamiltonian diffeomorphism on a closed surface is positive, then so is the topological entropy of any perturbation of it, as long as the perturbation is supported in a finite union of disks each with sufficiently small area. Joint work with Marcelo R.R. Alves and Beomjun Sohn.

Augmented Legendrian cobordant group

Yu Pan

For Legendrian links in the 1-jet space $J^1 M$, one can consider the equivalent class of Legendrian links up to Legendrian cobordant, introduced by Arnold, called Legendrian cobordant group. Augmentation is an algebraic invariant of Legendrian links coming from the SFT theory. It can be viewed as an algebraic analogue of the geometric information of Lagrangian fillings. Here we consider Legendrian links with augmentations as objects, called augmented Legendrian, up to augmented Legendrian cobordant. This also forms a group, called augmented Legendrian cobordant group. We compute this group explicitly for the case $M = \mathbb{R}$ or S^1 .

Contact spectral invariants and applications

Igor Uljarevic

In this talk, I will introduce a persistence module associated to a contact Hamiltonian and discuss numerical invariants extracted from it. This persistence module, based on contact Hamiltonian Floer homology by Merry-Uljarevic, can be constructed in the setting of a contact manifold with a strong weakly+ monotone filling. I will also discuss various applications of the theory, in particular, contact big fibre theorems, existence results of translated points, and sufficient conditions for orderability. This talk is based on joint work with Danijel Djordjevic and Jun Zhang.

Computational symplectic topology and RTBP

Otto van Koert

In this talk, we outline what kind of results in symplectic dynamics can be obtained using computational techniques. For concreteness, we focus on properties of symmetric orbits in the restricted three-body problem and the Birkhoff conjecture on the existence of a disk-like global surface of section. We give a brief outline of the technique, its advantages and its limitations.

Spinal open books and symplectic fillings with exotic fibers

Luya Wang

An evolving theme in symplectic topology has been the classification of symplectic manifolds, where pseudoholomorphic curves are often useful for establishing uniqueness and finiteness results. In the closed case, this traces back to Gromov and McDuff. In a series of works, Wendl used punctured pseudoholomorphic foliations to classify symplectic fillings of contact three-manifolds supported by planar open books, turning it into a problem about monodromy factorizations. In a joint work with Hyunki Min and Agniva Roy, we build on the recent works of Lisi--Van-Horn-Morris--Wendl in using spinal open books to further study the classification problem of symplectic fillings of higher genus open books. In particular, we provide the local model of the mysterious "exotic fibers" in a generalized version of Lefschetz fibrations, which captures a natural type of singularity at infinity. We will give some applications to classifying symplectic fillings via this new phenomenon.

Action-minimizing trajectories for Lagrangian flows

Jeff Xia

We study Lagrangian flows on manifolds with non-commutative fundamental groups. The action-minimizing trajectories or minimal measures can be defined in two different ways, in their homology classes or their homotopy classes. We use finite covering spaces to bridge these two concepts and uncover rich dynamical structures in these Lagrangian flows. The talk is based on joint works with Fang Wang.

RATIONAL SLICE GENUS BOUND AND MINIMAL GENUS PROBLEM

Jingling Yang

A fundamental problem in low-dimensional topology is to find the minimal genus of embedded surfaces in a 3-manifold or 4-manifold, in a given homology class. Ni and Wu solved a 3-dimensional minimal genus problem for rationally null-homologous knots. In this talk, we will discuss an analogous 4-dimensional minimal genus problem for rationally null-homologous knots. This is a joint work with Zhongtao Wu.

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